Write-Read Loop

Test-time Execution

Task input/ Instructions x_t

Agent/LLM

Working state w_t (local)

Committed output / Action y_t

Deployment Experience

- Trajectories
- Feedback
- User context
- Tool outcomes
- ...

Re-enter (C, G) Condition later decisions

Adaptive State

Write/ Update (A)

Retain
consolidate
revise
remove

Adaptive state z_t

Update z_t with new experience:

$$z_t \longrightarrow z_{t+1}$$

State can have different read interfaces

- **Context-Level** Textual / symbolic state
- **Representation-Level** Continuous internal state
- **Parameter-Level** Effective-parameter state

Three Read Interfaces

Context-Level

Read as explicit textual memory in prompt
reflections · rules · skills · episodes

Representation-Level

Read as latent conditioning in activations
soft tokens · retrieved KV memory · steering

Parameter-Level

Read as effective-parameter adaptation
parameter memories · adapters · weight updates

Three different re-entry points into later computation